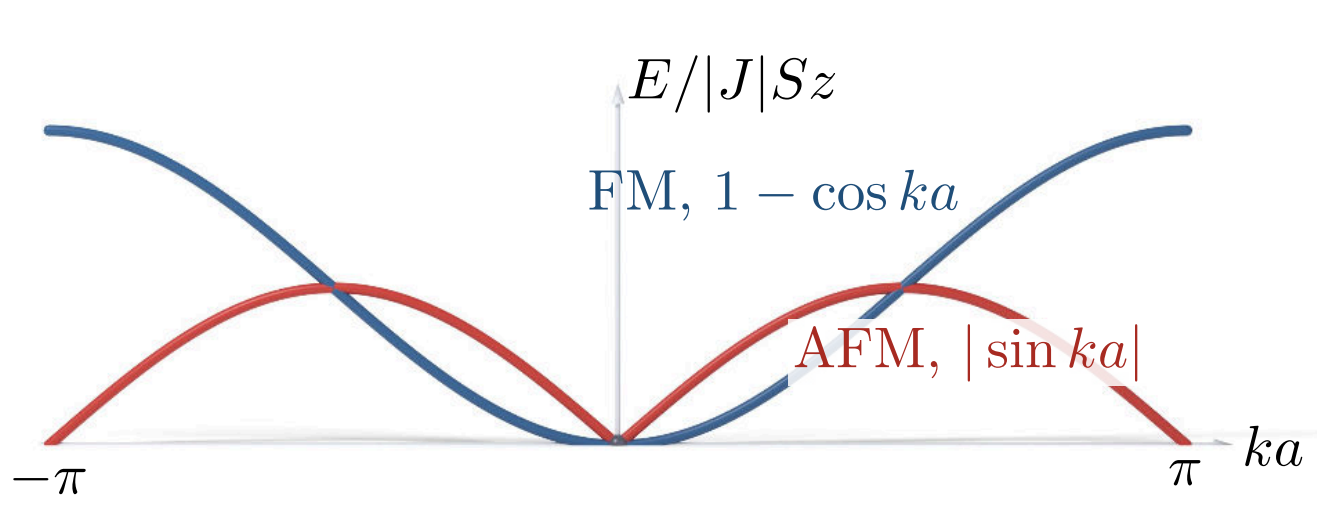
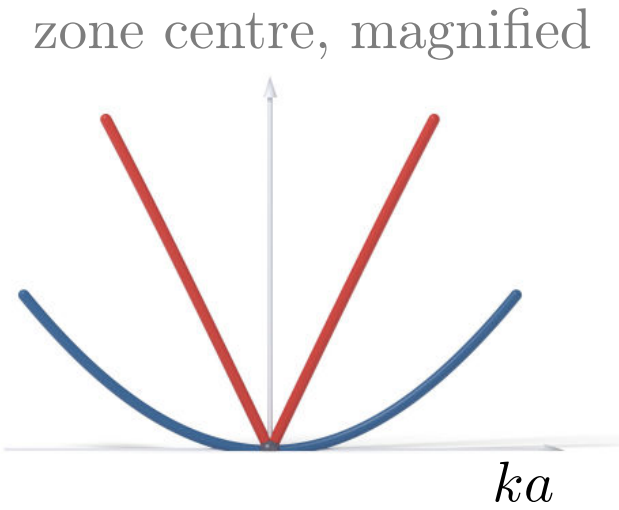

$E/|J|Sz$
FM, $1-\cos ka$
AFM, $|\sin ka|$
$-\pi$
π
ka


zone centre, magnified
ka